# PHYSICS

## WORKBOOK

## AND

## ACTIVITIES OF INTEGRATION

## SENIOR THREE




LOWER SECONDARY

NEW CURRICULUM

1ST EDITION

1st / Jan / 2024

**NEXT GENERATION LEARNER**

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# TABLE OF CONTENTS

Preface.....I

Acknowledgement.....II

Chapter 1.....1

Linear and non-linear motion.....2

Chapter 2.....38

Refraction, Dispersion and Colour.....39

Chapter 3.....59

Lenses and Optical Instruments.....60

Chapter 4.....80

General Wave Properties.....81

Chapter 5.....100

Sound Waves..... .....101

Chapter 6.....120

Heat Quantities and Vapours.....121

Chapter 7.....140

Stars and Galaxies.....141

Chapter 8.....160

Satellites and communication.....161

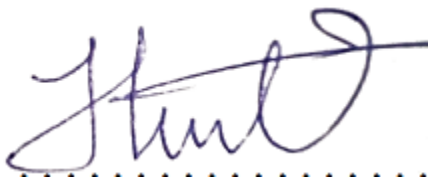
# Preface

This Learner's workbook has been written in line with the revised Physics syllabus for the new lower secondary curriculum. The revision questions, worksheets and activities of integration have been incorporated into this workbook. Through those activities of integration provided at the end of each chapter the learner is able to produce new knowledge, values and skills are required in the 21st century.

This has been done by providing a range of activities of integration which will enable the learner to research more through the internet in order to understand the applicability of knowledge learned at his or her respective school.

The learner is expected to be able to work as an individual, in pairs and groups according to the nature of the activities in order to be able to share learning experiences with their colleagues.

This Learner's workbook is one of the materials which are to be used to support the teaching and learning process of the new lower secondary curriculum.



**Mr. Onderi Kenneth**

Physics Teacher, Mandela secondary school Hoima.

# Acknowledgements

I would like to express a sincere appreciation to all those who worked tirelessly towards the production of this Learner's workbook.

First and foremost, I would like to thank my family and friends for supporting all my initiatives both financially and spiritually. my parents Mr. Omwandi Philemon, and Mrs. Akoth Rose Mary, my siblings Atosa Jonathan, Omwandi Philemon Junior, Adikini Esther lucy and Akoth Georgina. My friends Ashraf, Ntambi and Genza.

My gratitude also goes to the various institutions which provided staff who natured and supported me to become the physics teacher I am today. My thanks go to Mandela secondary school Hoima which provided the best environment to work from as a teacher, best reference books and best teachers in the profession like Mr. Centenary, Mr. Barongo, Madam Robina, Teacher Matthias, Mr. Mutattina, Mr. Akol, Teacher Gilbert, and Mr. Kagwa.

My thanks also go to Kakungulu memorial school Kibuli for best experienced colleagues like Mr. Hassan, Mr. Mukiibi, Mr. Abiib, Tr Gilbert, Mr. Kintu. They have been of great support towards these books.

Finally, I would like to acknowledge Greenhill Academy Kibuli for the good environment that enabled me to get used to the next generation teachers like Mr. Bemba, Mr. Amayo, Mr. Guma, Tr. Moses and Mr. Sunab.

I welcome any suggestions for improvement to continue making my service delivery better. Please get to me through kennethonderi@gmail.com or contact **0705476300** or **0771940855**

# Chapter 1

# Linear and Non-linear Motion



## Key learning outcomes

*By the end of this chapter, you should be able to:*

- ✓ Understand and apply relationship between speed, distance, and time.
- ✓ Understand the terms linear motion, speed, average speed, acceleration, and be able to investigate resistance to motion.
- ✓ Know and use the equations of motion.
- ✓ Understand the acceleration of bodies moving in a circular path and the effect of gravity and air resistance on moving bodies.
- ✓ Understand linear momentum and that it is conserved during collisions.
- ✓ Understand that momentum is conserved during collisions and the implications of this.
- ✓ Understand and apply newton's laws of motion.
- ✓ Understand the difference between vector and scalar quantities and give examples of each.
- ✓ Understand that several forces acting on the body can be represented by a single resultant force.

# Introduction

We can define motion as the change of position of an object with respect to time. A book falling off a table, water flowing from the tap, rattling windows, Walking, Running etc., all exhibit motion. Even the air that we breathe exhibits motion! Everything in the universe moves.

We live in a universe that is in continual motion. Every physical process in the universe is composed of motion of some sort. We might have noticed that different objects move differently. Some objects move in a curved path, some in a straight path and a few others in a different way that is why we have both linear and non-linear motion.

## WORKSHEET (SHORT ANSWER QUESTIONS)

## LINEAR MOTION



When an object moves in a straight line, this is known as linear motion. Think of the different times that you have seen some one, or something move in a perfectly straight line. When an object travels in a straight line it changes its position along with the time taken. There are several real-world examples of linear motion, such as: A bullet moving out of a gun. Falling of stone from a certain height, an athlete running 200 m on a straight track, train running in a straight track, car moving at a constant speed and many others.

1. Motion is mainly described in terms of the following terms in the table below. **Research** and match each term with its respective meaning.

| Term | Meaning |
|---|---|
| Speed | *is the rate of change of displacement moved with time* |
| Distance | This is the distance moved in a specified direction |
| Displacement | *is the rate of change of velocity with time* |
| Velocity | is the rate of change of the distance moved with time. |
| Acceleration | *is the length of path moved by a body* |

2. Research for the meaning of the following terms used in motion and match the following terms with their respective meaning.

| Term | Meaning |
|---|---|
| Initial velocity | ***is*** *the velocity of the body at any instant.* |
| Final velocity | is the ratio of the total displacement to the total time taken. |
| Instantaneous velocity | *Is when the rate of change of displacement with time is constant (doesn't change)* |
| Average velocity | is the velocity of the body at starting of motion. |
| Average speed | is when the rate of change of velocity with time is constant (doesn't change) |
| Uniform velocity | is the velocity of the body at ending point of motion. |
| Uniform acceleration | Is the total distance the object travels divided by the total time taken |

# Speed, distance and time and average speed

**SPEED** This is the distance covered per unit time.

$$speed = \frac{distance\ covered}{time\ taken}$$

The **SI unit** for speed is *metres per second (m/s or* $ms^{-1}$*)*

$$Average\ speed = \frac{Total\ distance\ covered}{Total\ time\ taken}$$

Another unit for speed used is Km/h or $kmh^{-1}$.

$$Average\ velocity = \frac{Total\ displacemenent}{Total\ time\ taken}\ \ or\ \ \left(\frac{v+u}{2}\right)$$

*Where* ***v*** *is the final velocity and* ***u*** *is the initial velocity*

## TRIAL NUMBERS (The longer the question, the more the difficulty)

3. A girl cycles for 3hrs at a speed of 40 km/h. What distance did she travel?



…… Answer = 120km

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4. A car travels a distance of 540km in 6 hours. What speed did it travel at?



…… Answer = 8 hours

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5. John is a runner. He runs the 100m sprint in 10.6s. What speed did he travel at? (in m/s) (**answer = 9.4 m/s**)



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6. A coach travels from the station to the beach, a distance of 576km away in 6hrs. The coach is only allowed to travel at a maximum speed of 90km/h. Did the coach break the speed limit? (**answer: yes, it travelled at 96km/h**)



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7. Marc was told his dinner would be ready at 18:00. He left his house at 12:00 and travelled in his car at an average speed of 75 km/h to his mum's house 500 km away. Did Marc make it home in time for dinner? At what minimum speed must he drive if he is to make it home in time for dinner?
(**Answers: No, he arrived at 18:40, he should drive at 83.3 km/h**)

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8. Callum writes down his jog times for each day.
   Mon – 15min  Tue – 10min  Wed – 12min  Thu - 5min  Fri – No jog.
   He jogs at a constant speed of 9km/h. Work out the distance he jogs each day. On which day did he jog the furthest? (Mon = 2.25km, Tue = 1.5km, Wed = 1.8km, Thu = 0.75km. he travelled farthest on Monday)



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   What have you learnt from the above scenario about relationship between time and distance at a constant speed?

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9. A body covers a distance of 10m in 4 seconds. It rests for 10 seconds and finally covers a distance of 90m in 60 seconds. Calculate the average speed
   …… **Answer = 5 m/s**
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10. A man runs 800m due North in 100 seconds, followed by 400m due South in 80 seconds. Calculate,
    (i) His average speed. (answer = 6.67 m/s)
    (ii) His average velocity. (answer = 2.22 m/s due North)
    (iii) His change in velocity for the whole journey (answer = 3m/s due North)
    (iv) Explain why the answers in (i) and (ii) are different.



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## ACCELERATION AND TICKER TAPE TIMER

Acceleration is the rate of change of velocity.

$$Acceleration\ (a) = \frac{change\ in\ velocity}{time} = \frac{V - U}{t}$$

$$change\ in\ velocity = final\ velocity(V) - initial\ velocity(U)$$

$$The\ S.I\ unit\ of\ acceleration\ is\ m/s^2\ or\ ms^{-2}$$

**Note** In motion when a body starts from **rest** its initial velocity (U) is zero and when body is brought to **rest** its final velocity (V) is zero

11. A car changes its velocity from **10ms$^{-1}$** to **50ms$^{-1}$** in **8s**. Find the acceleration of the car. (Answer; 5 ms$^{-2}$ )

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12. A body was accelerated at a rate of **6ms$^{-2}$** to acquire a velocity of **40ms$^{-1}$** in **4s**. Calculate the initial velocity of the body. (Answer: **16 ms$^{-1}$** )

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13. A car starts from **rest** and acquire a velocity of **60ms$^{-1}$** in **30s**. Find the acceleration of the car. (Answer: **2ms$^{-2}$** )

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14. A body moving with a velocity of **60ms$^{-1}$** is brought to rest in **12s**.
    (a) Find the acceleration of the car. (Answer: **-5 ms$^{-2}$** )
    (b) Why is the answer negative?
    (c) What physics term is used to refer to negative acceleration?

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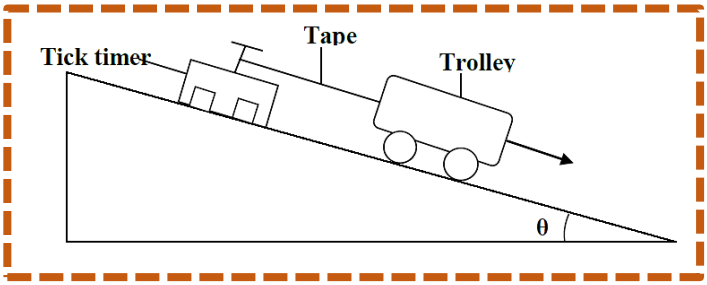
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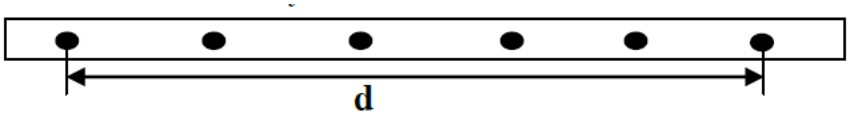
## TICKER TAPE TIMER




This is a device that is used to investigate velocity, speed, and acceleration of a body in a laboratory. It has an arm which vibrates regularly due to changing current of the A.C mains supply. As it vibrates it makes dots on the ticker tape.

### USEFUL FORMULARS IN CALCULATING VELOCITY USING TICKER TAPE TIMER.

Suppose the dots on the tape of the ticker time are as shown below.




$$velocity = \frac{Distance\ covered\ (d)}{Time\ taken\ (t)}$$

$$Time\ taken\ (t) = period \times number\ of\ spaces\ between\ the\ dots$$

$$period\ (T) = \frac{1}{frequency\ (f)} \quad or\ simply \quad T = \frac{1}{f}$$

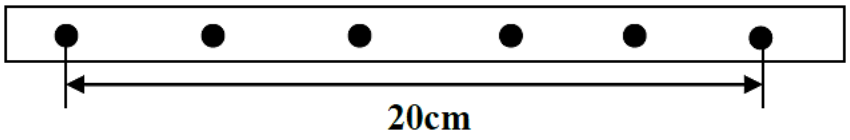
$period\ (T)$ *is the time taken to print any two successive dots on the tape*

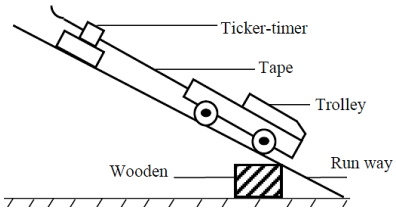
$frequency\ (f)$ *is the number of dots printed on the tape per second.*

$S.I\ unit\ of\ period\ is\ seconds\ (s)$

$S.I\ unit\ of\ frequency\ (f)\ is\ Hertz\ (Hz) or\ dots\ per\ second$

15. The ticker tape below was pulled by ticker timer. If its frequency is 50Hz. Calculate the speed at which the tape is pulled. (Answer: **2 m/s**)




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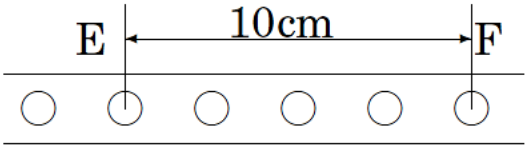
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16. The ticker tape shown in the figure below was pulled through a ticker timer which makes 50 dots per second.




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At what speed was the tape pulled?

(Answer: $1.25\ ms^{-1}$)

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**NOTE to calculate acceleration using ticker tape timer,**

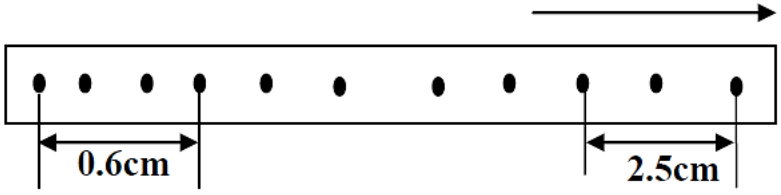
Note the initial distance $d_1$ in its initial time $t_1$ and determine the initial velocity, $u = \frac{d_1}{t_1}$ and note the final distance $d_2$ in its final time $t_2$ and determine the final velocity, $v = \frac{d_2}{t_2}$

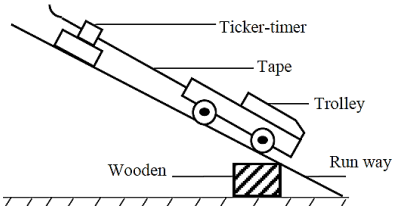
$$Acceleration\ (a) = \frac{change\ in\ velocity}{time\ taken} = \frac{v-u}{t} = \frac{\left[\frac{d_2}{t_2} - \frac{d_1}{t_1}\right]}{t}$$

$$Time\ taken\ (t) = period \times number\ of\ spaces\ between\ the\ dots$$

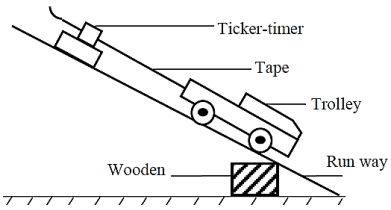
**N.B.** Spaces for acceleration are from the middle of initial distance to the middle of the final distance. This helps in obtaining time taken $t$

17. The figure below shows a piece of tape pulled through a ticker-timer by a trolley down an inclined plane. The frequency of the ticker-timer is 50Hz.




Calculate the acceleration of the trolley. (Answer $3.5\ ms^{-2}$)




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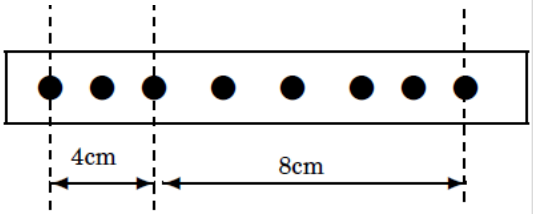
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18. Fig. 10 shows dots made on a ticker tape pulled by a trolley through a ticker-timer.




Describe the motion of the trolley if the frequency is 50Hz. (Is the trolley accelerating or decelerating? At what rate?)

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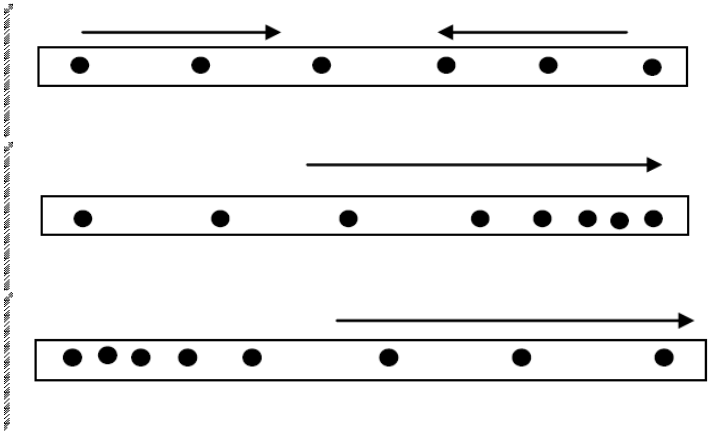
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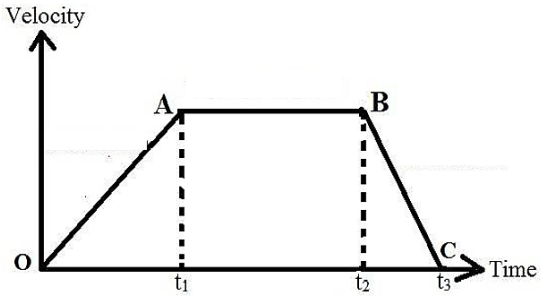
19. The figure below shows the tape of a ticker timer with dots printed on it. Identify the type of motion shown by the body in each of the figures.

(Select the right answer and fill in the table below. uniform acceleration, uniform deceleration, uniform velocity (no acceleration)

| Illustration | Description of motion |
|---|---|
|  | |

# GRAPHS OF LINEAR MOTION

20. The velocity-time graph below shows the motion of BODA BODA rider. He rides from home (point **O**) to school (point **C**) every day.




Along OA……

Along AB……

Along BC……

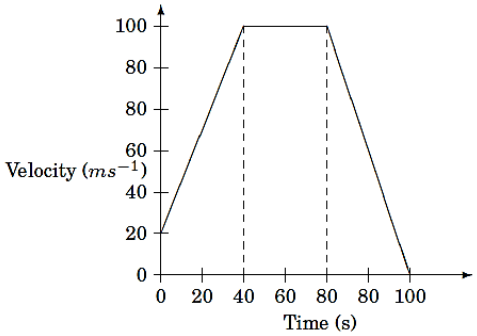
Indicate on the graph the region where he rode with:

(i) Uniform velocity

(ii) Uniform acceleration

(iii) Uniform deceleration (uniform retardation)

21. The velocity-time graph for a car is as shown in the figure below.




Find the total distance the car travels.

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22.a car traveling with a uniform velocity of **25 $ms^{-1}$** for **5 s** brakes and then comes to rest under a uniform deceleration in **8 s**.

(i) Sketch a velocity –time graph of the motion.

(ii) Find the total distance travelled. (answer **S = 225m**)

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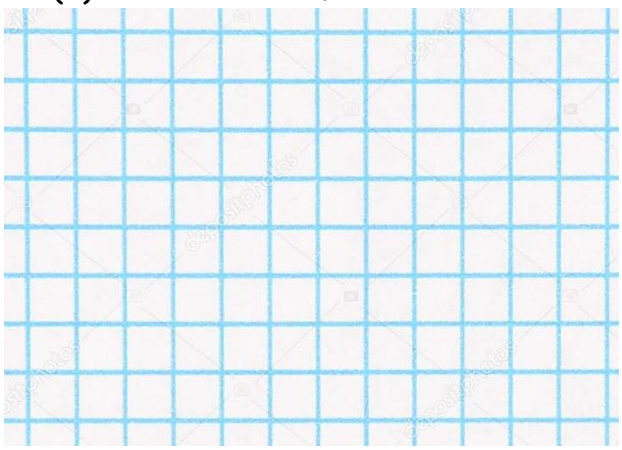
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23. Two vehicles **A** and **B** accelerate uniformly from rest. Vehicle **A** attains a maximum velocity of **30ms$^{-1}$** in **10s** while vehicle **B** attains a maximum velocity of **40ms$^{-1}$** at the same time. Both vehicles maintain these velocities for **6s**. They are then decelerated such that **A** comes to rest after **6s** while **B** comes to rest after **4s**.

(i) Sketch on the same axes a velocity-time graph for the motion of the vehicles.

(ii) Which of the two vehicles travelled more distance?



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24. The table below shows the variation of velocity with time for a body which has been thrown vertically upwards from the surface of a planet.

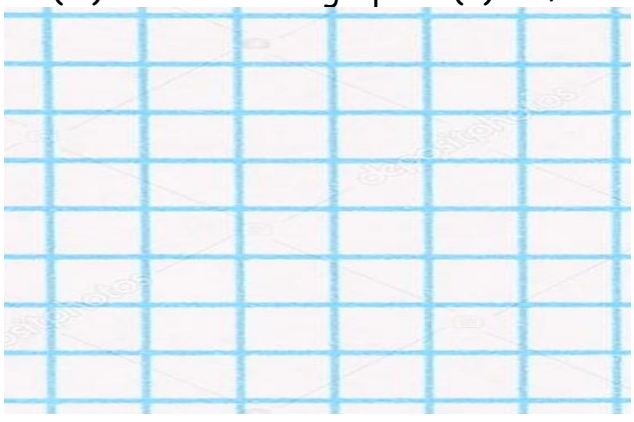
| Velocity(m/s) | 8 | 6 | 4 | 2 | 0 | -2 |
|---|---|---|---|---|---|---|
| Time(s) | 0 | 1 | 2 | 3 | 4 | 5 |

(i) What does the negative velocity mean?……

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(ii) Plot a graph of velocity against time.

(iii) Use the graph in (ii) to find the total distance travelled. (**Ans**: **17m**)



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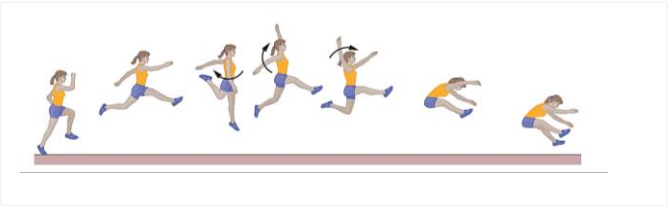
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# EQUATIONS OF MOTION

When a body starts with an initial velocity of **U** $\mathbf{ms^{-1}}$ and accelerated uniformly at a rate of **a** $\mathbf{ms^{-2}}$ to a acquire a final velocity of **V** $\mathbf{ms^{-1}}$ in time **t** s and covers a distance of **s m**, then the equations of motion of a body are.

| First Equation ($1^{st}$) | Second Equation ($2^{nd}$) | Third Equation ($3^{rd}$) |
|---|---|---|
| $v = u + at$ | $S = ut + \frac{1}{2}at^2$ | $v^2 = u^2 + 2as$ |

Where:

**u** is the initial velocity
**v** is the final velocity
**t** is the time taken
**a** is the acceleration.
**s** is the distance covered



**NOTE: deceleration (retardation) means negative acceleration (a is negative)**

25. A car started from **rest** it accelerates uniformly for **5s** at a rate of $\mathbf{4ms^{-2}}$. Calculate the final velocity. (Answer: **20 m/s**)

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26. A body traveling at $\mathbf{20ms^{-1}}$ is accelerated for **4s** at $\mathbf{5ms^{-2}}$. Calculate the average velocity. (Hint: $average\ velocity = \left(\frac{v+u}{2}\right)$ Answer: **30 m/s**)

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27. A body starts from **rest** and accelerates uniformly at $\mathbf{2ms^{-2}}$ for **3s**. Calculate the total distance travelled. (Answer: S = 9m)

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28. A body moving at **40** $\mathbf{ms^{-1}}$ decelerates uniformly for **20s** at $\mathbf{3ms^{-2}}$. Calculate distance covered. (Answer: **S = 200 m**)

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29. Calculate the final (maximum) velocity of a body traveling at **4 $ms^{-1}$**. When it accelerates at **$2ms^{-2}$** and covers a distance of **5m**. (Answer: **V = 6 $ms^{-1}$**)

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30. A body traveling at **90km/hr** is retarded to rest at **$20ms^{-2}$**. Calculate the distance covered. (Answer: **S = 15.6 m**)

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31. A jumbo jet must reach a speed of **360 km/h** on the runway for take-off. What is the lowest constant acceleration needed for take-off from a **1.80 km** runwau? (Answer: **a = 2.78 $ms^{-2}$**)

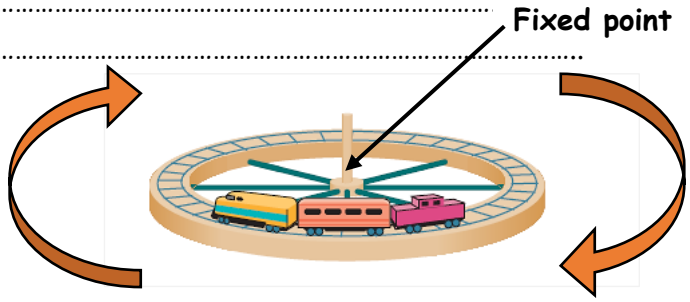


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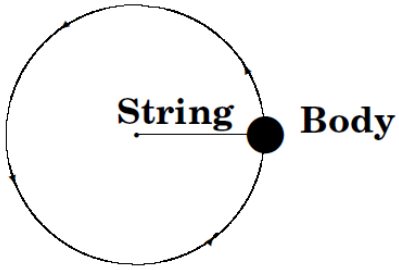



## CIRCULAR MOTION

Circular motion is motion in which a body moves in a circle about a fixed point. Examples include.

(i) Rotation of the earth
(ii) Motion of electrons around the nucleus

32. A body attached to a string is swung in a vertical circular path in air as shown in below.




Copy the above diagram and on it indicate and name all the forces acting on the body if the body is moving in an anti-clockwise direction. (*centripetal force, centrifugal force, weight, applied tangential force, tensional force*)

33. What are the applications of circular motion? Name any four.

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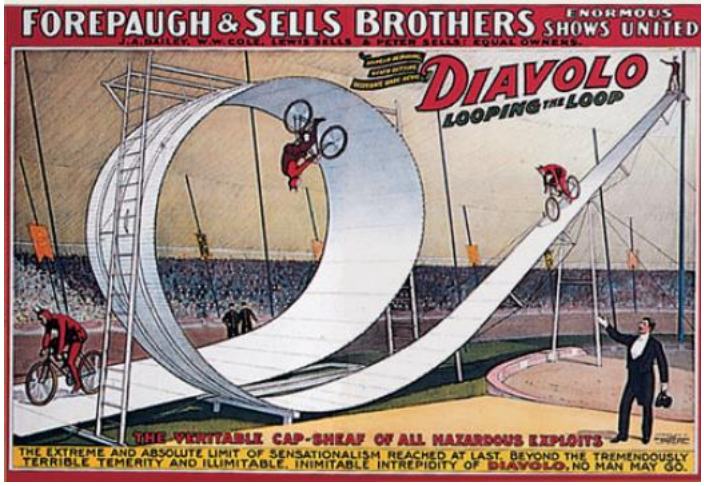
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## Did you know?

- ✓ Water in a bucket can be whirled in a vertical circle without any spills if it is done with a high enough speed? Try it out after class and research on the possible reasons why? (precaution: it spills at low speed of whirling so be warned)
- ✓ The acceleration of a body moving in a circular path is called **centripetal acceleration**. It is always directed towards the centre of the circle.
- ✓ The force that keeps objects moving in a circular path is called **centripetal force** and is always acts towards the centre of the circle.

34. Vertical circular motion is just like magic. Such motion seems to defy the gravitational force. In a 1901 circus performance, Allo "**Dare Devil**" Diavolo introduced the stunt of riding a bicycle in a loop the-loop as shown in the figure below. At one point, the rider's head is facing down but surprisingly he continues riding safely.




*Explain why the rider at the top of stunt will not fall if he is riding fast enough? And why does he fall off at slow speeds?*

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# RESISTANCE TO MOTION

## The effect of gravity air resistance and water resistance on moving bodies



Air resistance and gravity are the two fixed forces of nature which move on any object on Earth. Water resistance is only experienced by bodies moving through water. For example, when you drop an egg in a saucepan filled with water it falls slowly to the bottom of the saucepan without breaking.

- **Gravity** - the force that brings things to the ground. E.g., a pen falls from the table due to gravity, leaves fall due to gravity.
- **Air resistance** – is the friction between air and an object.
- **Water-resistance** – is the friction between water and an object.



**Air resistance** is a kind of friction (a force that opposes motion) which occurs between air and another object. It is the force that the object experiences as it passes through the air.

### Real life examples

- ✓ When wind blows, it is more difficult to walk, or run. This is because the air resists your force and pushes back against you.
- ✓ The wind slows us while we're riding bikes.
- ✓ feathers or light papers fall slowly when wind blows due to air resistance.
- ✓ On a windy day, when you see leaves falling from a tree really slowly, that's because air resistance is slowing down its fall.
- ✓ When a skydiver jumps out of an airplane, they open a parachute. Air resistance causes them to parachute slowly to the ground.

## Air Resistance and Streamlining





For things to move quickly and efficiently through air or water, objects need to have a small surface area. This is because the bigger the surface area, the greater the resistance. This means that objects need to be **streamlined** in order to push against the air's force.

**A streamlined body** is the one which is pointed at the ends and broad at the middle.



35.Explain why sports cars and fighter jets are streamlined?

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36.Explain why it is more difficult to ride while putting on a light large shirt?

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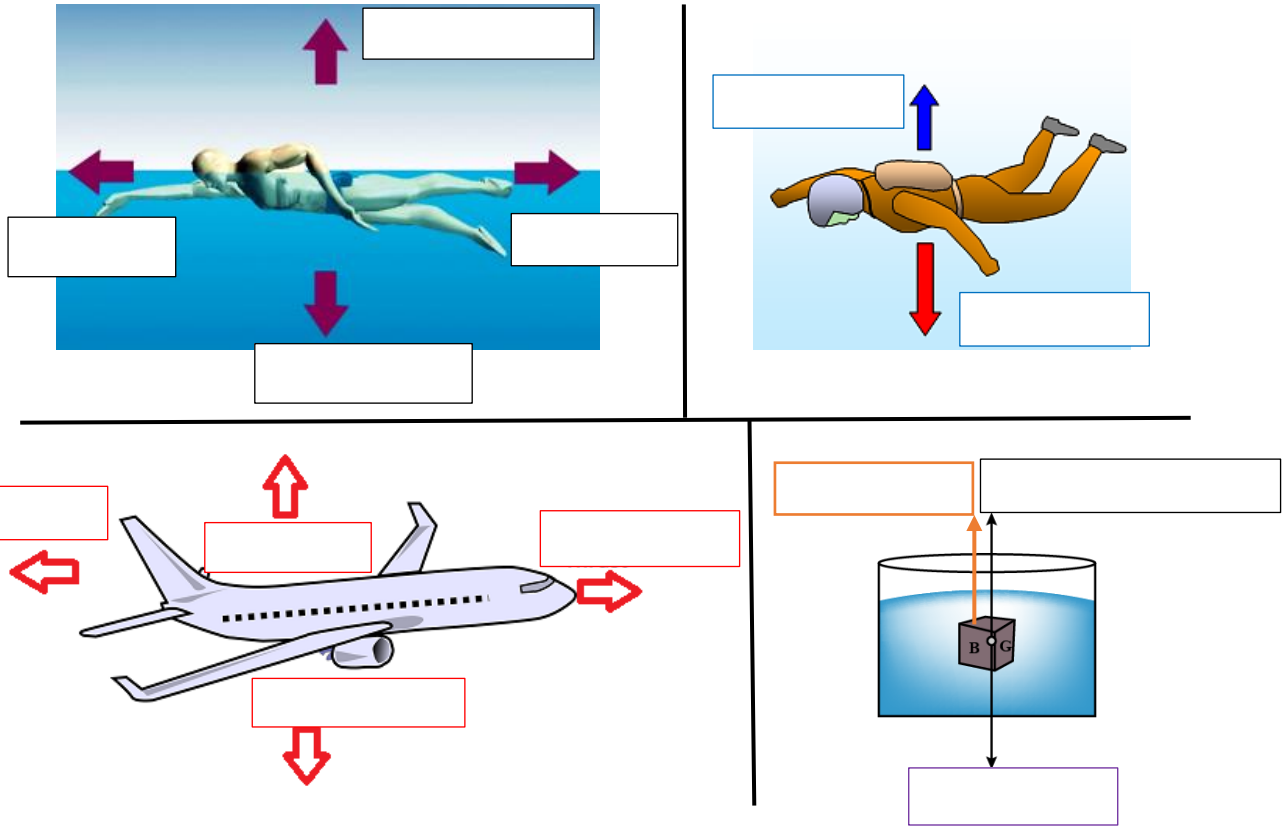
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**Water resistance** is only experienced by bodies moving through water. For example, when you drop an egg in a saucepan filled with water it falls slowly to the bottom of the saucepan without breaking. It is more difficult to run through water.

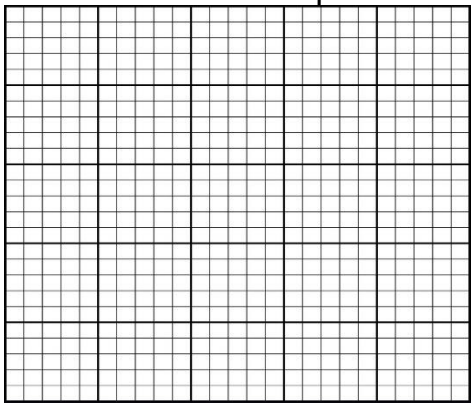
## MOTION OF A BODY FALLING IN A VISCOUS FLUID

A fluid is a substance which can flow e.g., liquids and gases.

37. When a body moves through a fluid, it can be acted upon by several forces: (Upthrust force, viscous drag, weight, lift force, Thrust). Label the forces acting on this body shown below.



38. Sketch a velocity time graph for a body that falls freely through a viscous fluid and use it to explain the motion of a body through a viscous fluid.



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# FREELY FALLING BODY (MOTION UNDER GRAVITY)

Motion under gravity is the motion described by a body falling freely, where the only force acting on it is the gravitation force. The acceleration of such a body is called ***acceleration due to gravity***.

**Acceleration due to gravity is the rate of change of velocity for freely falling body under the force of gravity.**

- ✓ When a body is falling vertically **downwards** under the force of gravity its velocity keeps on increasing and acceleration due to gravity acting is positive ($a = g = 10\ ms^{-2}$ on planet earth)
- ✓ When a body is moving vertically **upwards** under the force of gravity, its velocity keeps on reducing until it reaches maximum height when the velocity is zero. The acceleration due to gravity is negative ($a = -g = -10ms^{-2}$)

## Equations of motion For a body in free fall

| S/N | UPWARD MOTION ($a = -g$) | DOWNWARD MOTION ($a = g$) |
|---|---|---|
| 1st | $v = u - gt$ | $v = u + gt$ |
| 2nd | $S = ut - \frac{1}{2}gt^2$ | $S = ut + \frac{1}{2}gt^2$ |
| 3rd | $v^2 = u^2 - 2gs$ | $v^2 = u^2 + 2gs$ |

**NOTE**: The velocity of a body at maximum height above the ground is zero because at maximum height the body is at rest (not moving).

39.a) With what speed must a ball be thrown vertically from ground level to rise to a maximum height of **50 m**? (take **g = 10 ms$^{-2}$**(Answer: **u = 31.6 m/s**)
(b) How long will it be in the air? (Answer: **t = 6.3s**)

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40. Raindrops fall **1700 m** from a cloud to the ground.
    (a) If they were not slowed by air resistance, how fast would the drops be moving when they struck the ground? (Answer: speed is **184.4 m/s** which is in range with the speed of a bullet in most guns)
    (b) Would it be safe to walk outside during a rainstorm?

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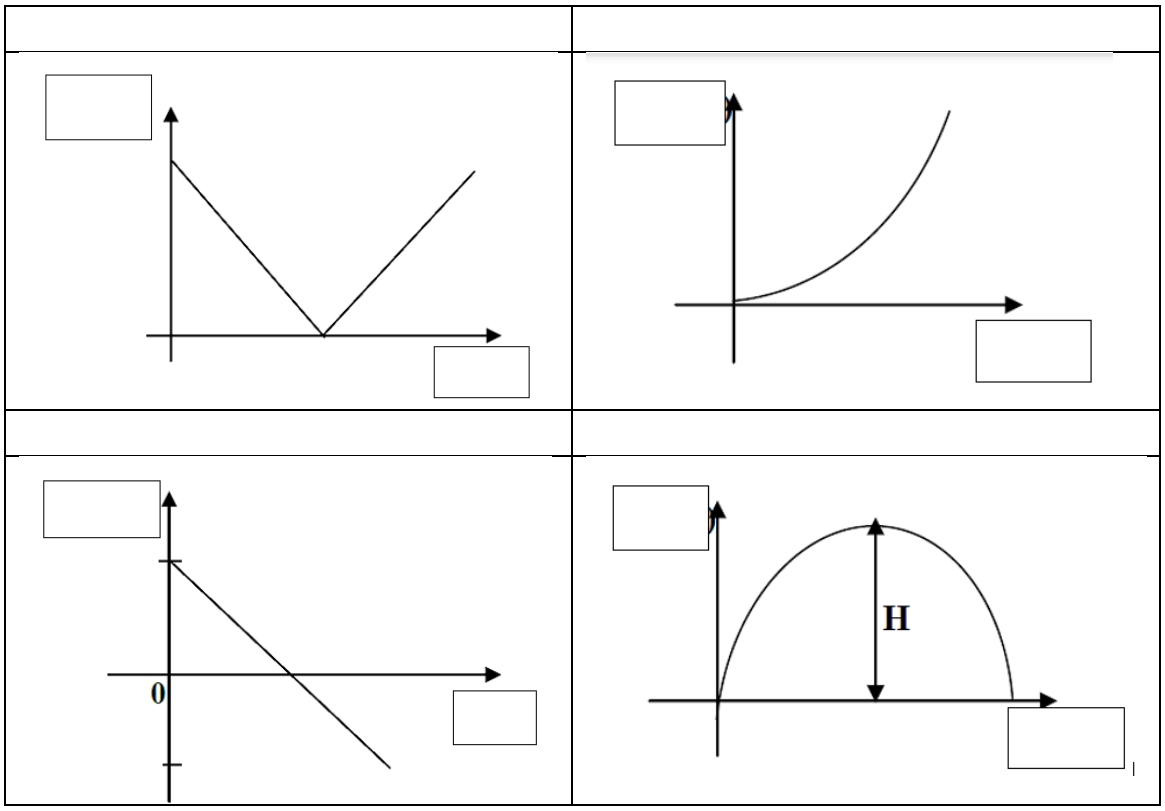
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41. A small iron ball dropped from the top of a vertical cliff takes **2.5s** to reach the bottom of the cliff. Find
    (i) The speed with which it strikes the bottom. (Answer: **V = 25 $ms^{-1}$**)
    (ii) The height of the cliff. (Answer: height is **31.25 m**)

42. When a body is thrown vertically upwards the following graphs describe the motion of the body.

| | |
|---|---|
| Distance – time graph | Displacement – time graph |
| Speed – time graph | Velocity – time graph |

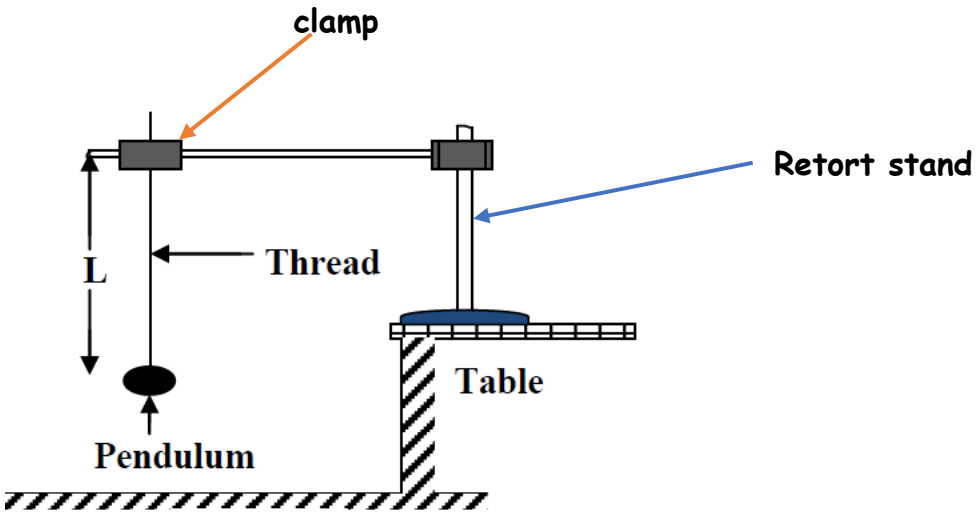
Identify the right name for the graphs shown in the table below from the table shown above and also label the axis of each of the graphs you named.



# An experiment to determine acceleration due to gravity using a simple pendulum.

**Apparatus**: a thread, a bob, clock, clamp, metre ruler.

**Diagram**; the apparatus is arranged as shown below




## PROCEDURES (STEPS FOLLOWED IN THE LAB)

a) Suspend the pendulum bob from a retort stand using a piece thread provided as shown in the figure above.

b) Start by measuring length, $l = 90cm$ of the thread from the midpoint of the bod to the clamp.

c) Displace the bob through a small angle and release it to oscillate.

d) Measure and record the time, **t**, for **20** oscillations.

**e)** Calculate the period **T** (time for 1 oscillation) $T = \frac{t}{20}$

f) Repeat procedures (b) to (e) for different lengths of the pendulum. ( $l = 80, 70, 60, 50, 40, and\ 30cm$ )

g) Tabulate your results including values of $T^2$

h) Plot a graph of $T^2\ against\ l$.

i) Determine the slope, **S** of the graph.

j) Calculate the acceleration due to gravity, from the expression $g = \frac{4\pi^2}{S}$

**The workspace below should be filled by the student after carrying out the experiment in the laboratory.**

## NOTE:

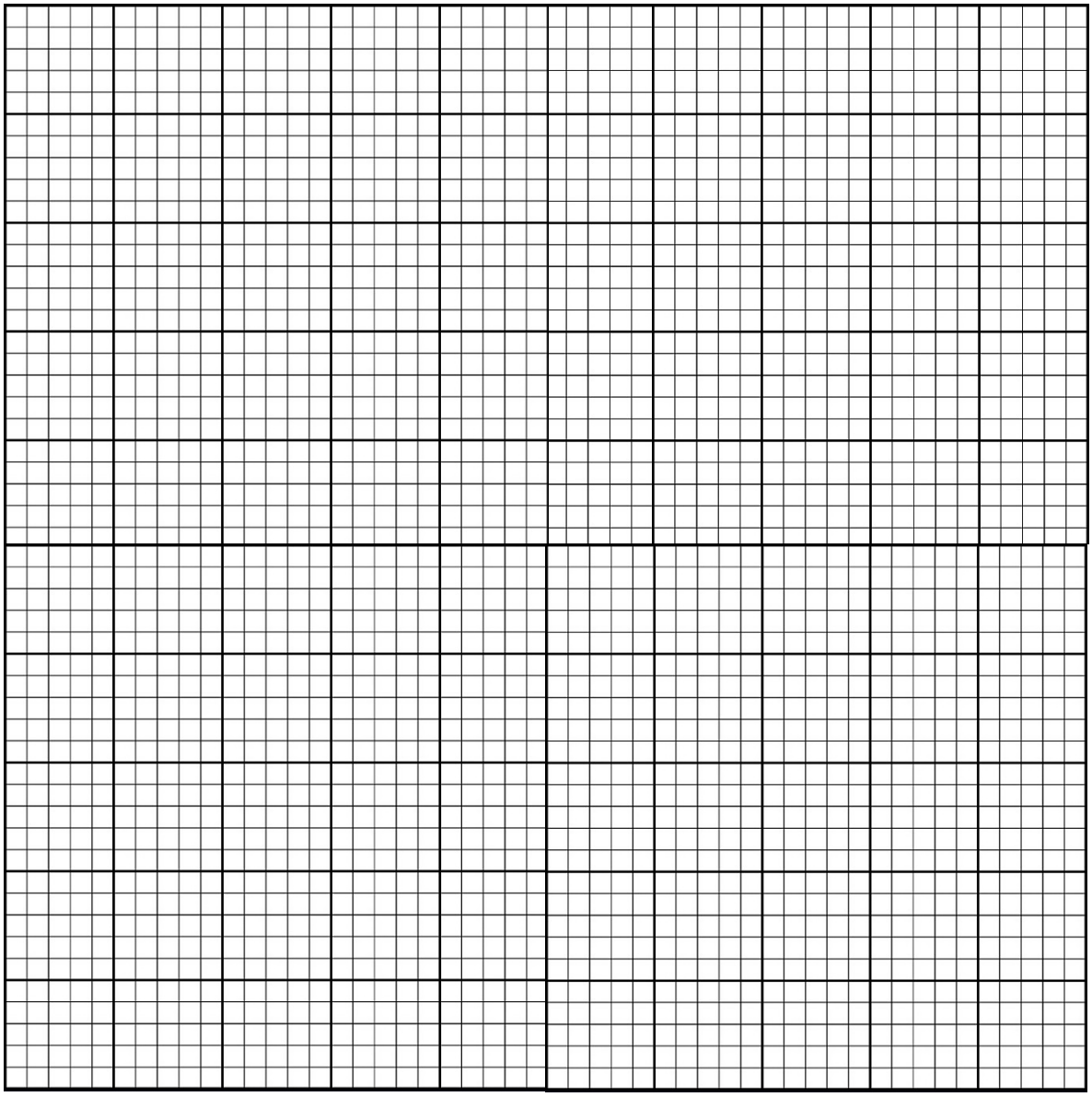
- ✓ The student should know all the procedures followed because they are not provided in examinations.
- ✓ The student may use his or her own values of length for example. $l = 80, 75, 60, 55, 50, 45, and\ 40cm$
- ✓ The student may also his or her own periodic time e.g. $T = \frac{t}{10}$ and $t$ is time for 10 oscillations

## TABLE OF RESULTS

| $l(cm)$ | $l(m)$ | $t(s)$ | $T(s)$ | $T^2(s^2)$ |
|---|---|---|---|---|
| | | | | |
| | | | | |
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Where $T = \frac{t}{20}$ and $t$ is time for 20 oscillations

## A graph of $T^2$ *against* $l$



Calculate the slope, 'S' of the graph.

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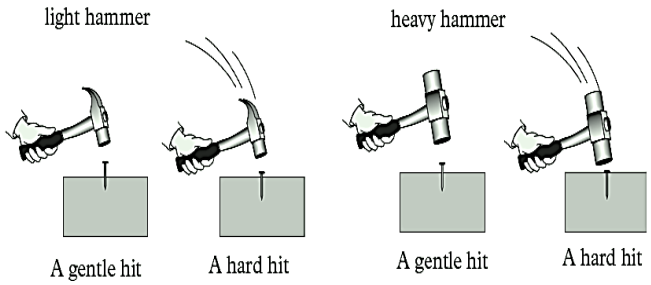
calculate the acceleration due to gravity using the expression.

$g = \frac{4\pi^2}{S}$

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# MOMENTUM

This is the product of mass of the body and its velocity.

$$Momentun = mass \times velocity$$

$$The\ S.I\ unit\ of\ momentum\ is\ kgms^{-1}$$

43. The momentum of a body depends on its mass and velocity.
   - ✓ The momentum of a body increases with increase in its mass.
   - ✓ The momentum of a body increases with increase in its velocity.

   Using the above knowledge, explain the following real-life situations.

   (i) A heavy hammer can drive a nail deeper into a piece of wood than a lighter hammer.

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   (ii) A heavy footballer is harder to stop once in motion than a small footballer.

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   (iii) A fast-moving ball is not easier to stop than a slow-moving ball.

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   (iv) A fast-moving car causes more damage when it makes an accident than a slow-moving car.

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44. A car of mass **1000kg** travels with a velocity of **50ms$^{-1}$**. What is the momentum of the car? (Answer: **50,000 kgms$^{-1}$**)

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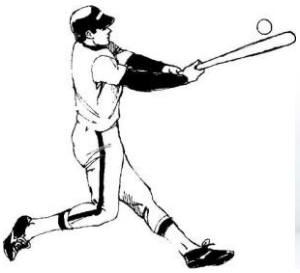
45. An object has a mass of **200kg** and a momentum of **3800kgms$^{-1}$**. At what velocity is it moving? (Answer: **19 m/s**)

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# Collisions







This is when two or more bodies come into contact.for example. An interaction between two billiard balls, an object falling on the floor, and a bat hitting a baseball and others. Name any other two real life cases where collision takes place.

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All collisions follow the **law of conservation of momentum**,

## The principle of conservation of linear momentum:

It states that when two or more bodies collide their total linear momentum remains constant provided no external force is acting.

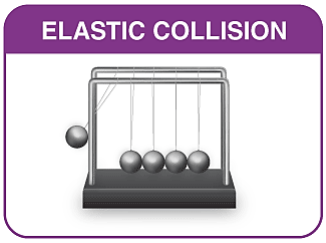
In simple words total linear momentum of the colliding bodies **before** collision is **equal** to total linear momentum **after** collision if no external forces acts.

$$Total\ momentum\ before\ collision = Total\ momentum\ after\ collision$$

There are two **types** of collision.

i) Elastic collision.

ii) Inelastic collision





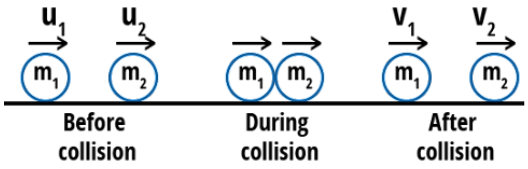
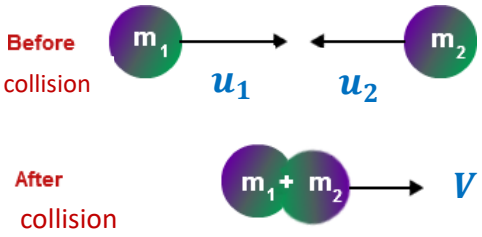


### Elastic collision

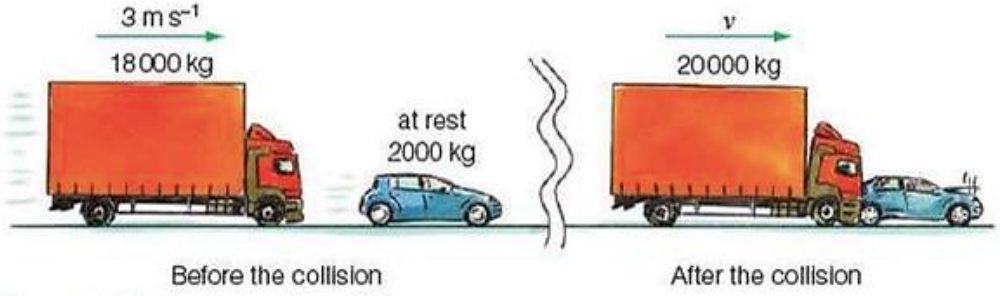
This is a type of collision where the colliding bodies separate after collision and move with different velocities.

### Inelastic collision

This is a type of collision where the colliding bodies stay together after collision and move with the same velocity.

| Elastic collision | Inelastic collision |
|---|---|
| Bodies separate after collision.<br>  <br> | Bodies stick together after collision.<br>  <br> |
| Bodies move with **different** velocity after collision. | Bodies move with **same** velocity after collision |
| Momentum is conserved.<br>$m_1u_1 + m_2u_2 = m_1V_1 + m_2V_2$ | Momentum is conserved.<br>$m_1u_1 + m_2u_2 = (m_1 + m_2)V$ |
| Kinetic energy is conserved.<br>$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1V_1^2 + \frac{1}{2}m_2V_2^2$ | Kinetic energy is not conserved |
|  |  |

46. The following image shows an unfortunate situation: a van fails to stop as it approaches a line of traffic and hits a stationary car; and they move forwards together - is this elastic or inelastic collision? Prove it **mathematically**. How can collision and its effects be minimized?




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47. An object of mass **2kg** moving at **5 $ms^{-1}$**, collides with another object of mass **3kg** which is at rest. Find the velocity of the two bodies if they stick together after collision. (Answer: **2m/s**)

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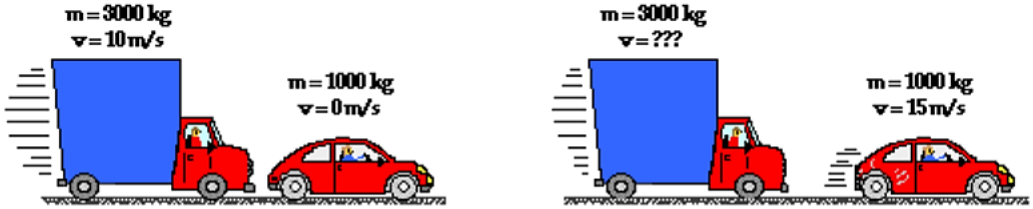
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48. A **3000 kg** truck moving with a velocity of **10 m/s** hits a **1000 kg** parked car. The impact causes the **1000 kg** car to be set in motion at **15 m/s**. Assuming that momentum is conserved during the collision, determine the velocity of the truck immediately after the collision. (Answer: **5 m/s**)

**BEFORE** **AFTER**




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49. A body of mass **4.0kg** moving with a velocity of **25$ms^{-1}$** collided with another body of mass **6.0kg** moving with a velocity of **40$ms^{-1}$** from the **opposite** direction. If **4.0kg** mass moves with a velocity of **10$ms^{-1}$** after collision, find the velocity of **5.0kg** mass after collision.

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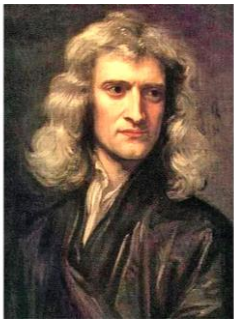
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# NEWTON'S LAWS OF MOTION

**Newton's laws of motion** are three basic laws of classical mechanics that describe the relationship between the motion of an object and the forces acting on it. These laws were first formulated in **1687 (17th century)** by English physicist and mathematician Isaac Newton who is considered by many as the father of modern science.



We have about three Newton's law of motion, includes.

i. Newton's first law of motion (law of inertia)
ii. Newton's second law of motion (F = ma)
iii. Newton's third law of motion (law of action and reaction)

## First law of motion (law of inertia)

It states that everybody continues in its state of rest or uniform motion in a straight line unless acted upon by an external force. This law is sometimes referred to as **the law of Inertia**.

Inertia is the tendency of a body to remain at rest or keep moving with a uniform velocity. **OR** is the reluctance of a body to start moving or stop moving.

50. One Saturday evening john's parents sent him to buy salt in the nearby shop. He then quickly picked his bicycle and rode to the shop. he was riding at a high speed and suddenly he saw a snake crossing the road, he got frightened and applied brakes immediately which made him to fall off the bicycle and got injured in the process.
    (a) Why did john fall off the bicycle when he suddenly applied brakes?
    (b) Which of the newton's laws of motion was in play here?
    (c) What lessons do you learn from john's story about motion?
    (d) What safety precautions can be taken to reduce injuries especially in public means of transport like Buses, Taxis and Boda Boda.
    (e) Name and briefly describe any other two real life situations where you have experienced this law of inertia (1st law)

    a)……
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    b)……
    c)……
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d)……

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e)……

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51. Describe and explain what happens to you when you are in a school bus going for a tour and suddenly the driver:
    (a) Applies brakes.
    (b) Increases the speed (accelerates)

(a)……

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(b)……

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## Second law of motion ☹ (F=ma)

It is states that the rate of change of momentum of a body is directly proportional to the force applied and takes place in the direction of force.

$Force\ \alpha\ rate\ of\ change\ of\ momentum$

$$F\ \alpha\ \frac{change\ of\ momentum}{time}$$

$change\ of\ momentum = final\ momentum - initail\ momentum = \mathbf{mv - mu}$

$$F\ \alpha\ \left(\frac{mv - mu}{t}\right)$$

$$F\ \alpha\ m\left(\frac{v - u}{t}\right)\ \ but\ \ \mathbf{a} = \left(\frac{\mathbf{v - u}}{\mathbf{t}}\right)$$

Where, **m** = mass, **F**= force, **v** = final velocity, **u** = initial velocity, **t** = time, **a** = acceleration

$\mathbf{F\ \alpha\ ma}$ *on removing proportionality sign* $(\alpha)$ $\boldsymbol{F = kma}$ *where* $\boldsymbol{k}$ *is a constant*

$if\ k = 1\ \ then\ \boldsymbol{F = ma}$

The SI unit of force is a **Newton (N)**.

A **Newton** is the force which gives a mass of $1kg$ an acceleration of $1ms^{-2}$.

52. An object of mass $4kg$ accelerates at a rate of $5ms^{-2}$. Calculate the resultant force acting on it. (Answer: **F = 20N**)

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53. Calculate the acceleration produced by a force of **25N** on an object of mass **2 tonnes**. (Answer: **a = 0.0125 $ms^{-2}$**)

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54. In Newton's second law, (**F = ma**) acceleration depends on two variables i.e. the **net force** acting on the body and the **mass** of the body i.e.. $a = \frac{F}{m}$

- ✓ This means an **increase** in force **increases** the acceleration of the body.
- ✓ and an **increase** in mass **reduces** the acceleration of the body.

**Use the above knowledge to explain the following real-life situations.**

(a) When in a supermarket, it's easier to push an empty cart than to push a loaded one.

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(b) Among two people walking, if one is heavier than the other one, the heavier person will walk slows than the lighter person.

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(c) In formula one racing, the mass of the cars is kept as low as possible.

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(d) Its easier to push a loaded wheelbarrow when your brother pulls it using a rope as you are pushing it forward.

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# Impulse and change in momentum of the body.

In a collision, an object experiences a force for a given amount of time that results in its mass undergoing a change in velocity (i.e., that results in a momentum change).

**Impulse (I)** is the product of force and time for which it acts.

$$Impulse(I) = Force \times time \quad or \quad simply \quad I = Ft$$

$$F = ma = m\left(\frac{v - u}{t}\right) \; since \; a = \left(\frac{v - u}{t}\right)$$

$$Ft = m(v - u) = mv - mu = change\ in\ momentum = Impulse\ (I)$$

Therefore, **Impulse (I)** is the change in momentum of the body.

Deduce the S.I unit of impulse from the following.

(i) $I = Ft$ ……

(ii) $I = mv - mu$ ……

55. A body of mass $2kg$ changes its velocity from $10ms^{-1}$ to $45ms^{-1}$ after a period of time. Calculate the impulse on the body. (Answer: 70 kgm/s)
56. A body of mass **4.5kg** accelerates uniformly at $2ms^{-2}$ for 5 seconds. Calculate the impulse on the body. (Answer: 45 Ns)
57. **First, we will examine the importance of the collision time in affecting the amount of force that an object experiences during a collision.**

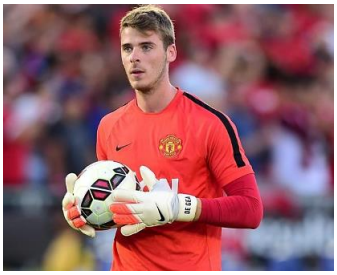
Combinations of Force and Time Required to Produce 100 units of Impulse

| Force | Time | Impulse |
|---|---|---|
| 100 | 1 | 100 |
| 50 | 2 | 100 |
| 25 | 4 | 100 |
| 10 | 10 | 100 |
| 4 | 25 | 100 |
| 2 | 50 | 100 |
| 1 | 100 | 100 |
| 0.1 | 1000 | 100 |

**INSTRUCTION**. Use the table above to help you answer the questions below. Select one of two answers provided in brackets and fill in the blank spaces.

a) We observe that the greater the time over which the collision occurs, the …… the force acting upon the object. (greater, smaller)
b) Thus, to minimize the effect of the force on an object involved in a collision, the time must be……(reduced, increased)
c) And to maximize the effect of the force on an object involved in a collision, the time must be…… (reduced, increased)

# Application of Momentum and Impulse in Our Daily Life





58.**Research** and explain the following daily life scenarios.

| | |
|---|---|
| ✚ A goalkeeper draws his hands towards his body when catching a fast-moving ball.<br>✚ Goal keepers wear soft gloves that absorb shocks on their hands. | **Explanation**<br>……<br>……<br>……<br>……<br>…… |
| ❖ A person jumping from a high ground to the floor bends his knees upon making contact otherwise his/she leg break.<br>❖ High jumpers and long jumpers land on sand or soft cushions | **Explanation**<br>……<br>……<br>……<br>……<br>…… |
| ✓ Glassware is wrapped in soft boards while packing.<br>✓ Objects that easily break like eggs are packed in soft, shock-absorbing boxes. | **Explanation**<br>……<br>……<br>……<br>……<br>…… |



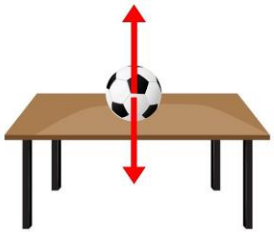

*A long jumper bends his knees as he lands on sand during Olympic games as shown in the figure above.*

# Third law☹ (Action and Reaction law)




It states that every action there is always an equal but opposite reaction.

59. A ball is placed on the table as shown below. Label the forces acting.



(a) Briefly explain the above diagram using newton's 3rd law of motion

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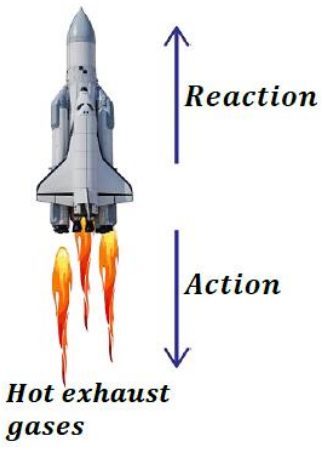
(b) Name any other applications of newton's 3rd law besides the above.

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## Application of the principle of conservation of momentum & 3rd law

60. The principle of conservation of momentum is applied in the following cases shown in the pictures below, **research** and make notes on how it is applied in each case.

| Rocket propulsion | Explanation |
|---|---|
|   | …… …… …… …… …… …… …… …… …… |

## Jet engine propulsion




## Explanation

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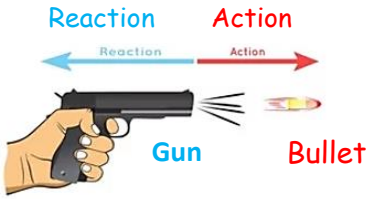
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## Recoil of a gun




## Explanation

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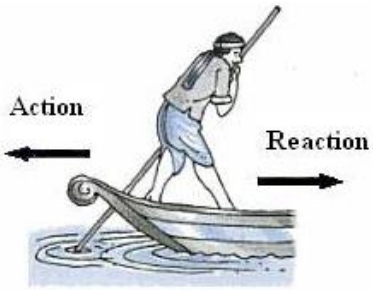
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## Rowing of a boat




## Explanation

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**61.** A bullet of mass **200g** is fired from a gun of mass **4kg**. If the muzzle velocity of the bullet is **400ms$^{-1}$**, calculate the recoil velocity. (Ans; **-20 m/s)**

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# VECTOR AND SCALAR QUANTITIES

A physical quantity is any physical property of a body that can be measured, and they include, area, volume, time, mass, distance, speed, temperature, velocity, acceleration, pressure, force, momentum and many others.

Physical quantities are divided into two categories namely:

(i) Scalar quantities
(ii) Vector quantities

62. *Research for the meaning of scalar quantities and vector quantities and list atleast five examples each. What are the differences between scalar and vector quantities.*

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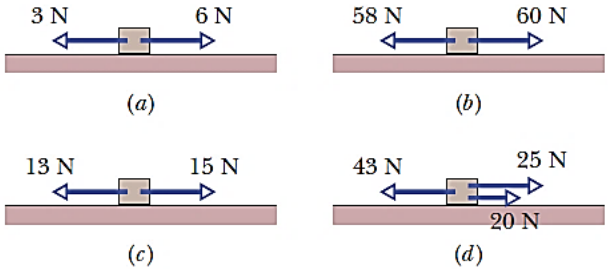
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# RESULTANT FORCE (NET FORCE)

Learners should be able to understand that a number of forces acting on a body can be represented by a single resultant force.
The sum of all forces acting on an object is called the resultant force OR net force.

- If the forces are acting in <u>opposite</u> <u>direction</u>, they are <u>subtracted</u> for example if two people are pulling a rope in opposite directions, the resultant force is the difference between the two forces.
- If the forces are acting in the <u>same</u> <u>direction</u>, they are <u>added</u>. For example, 10 people pushing a car out of a trench.

63. The figure below shows the same breadbox of mass **5kg** in four situations where horizontal forces are applied.




*Rank the situations according to the magnitude of the box's acceleration, greatest first. (show all your working clearly)*

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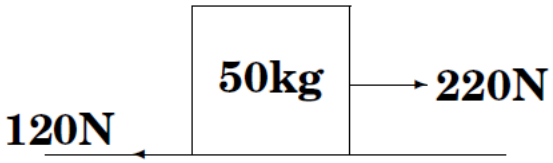
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64. A block of mass **50*kg*** is pulled from <u>rest</u> along a horizontal surface by a rope tied to one face of the block as shown in the figure below.




(i) Find the resultant force acting on the block.
(ii) Find the acceleration of the block.
(iii) Calculate the distance moved by the block in 4.0*s*.

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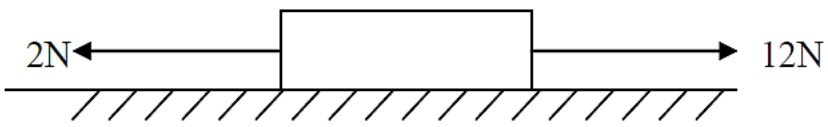
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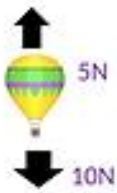
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65. The figure below shows a wooded block of mass 4.0kg acted upon by forces of 12N and 2N.




Calculate the acceleration of the block (Answer: **2.5 $ms^{-2}$**)

66. The figure below shows a hot air balloon being acted upon by a number of forces. Determine the resultant force on each balloon. (Include direction)




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$R.F =$




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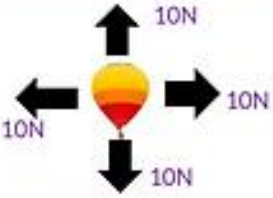

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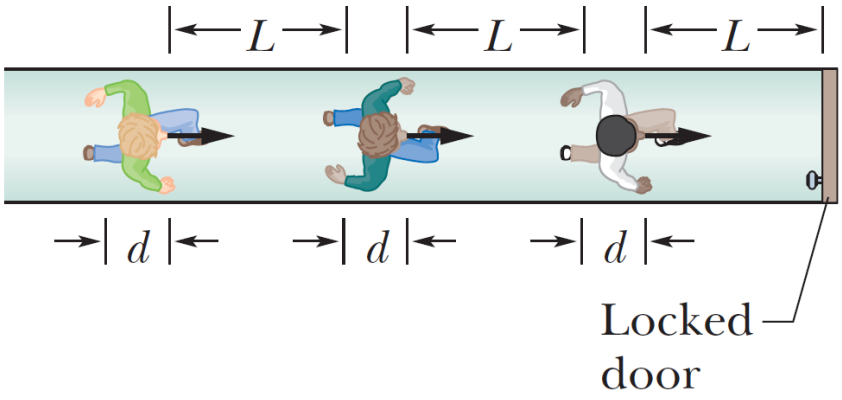



$R.F =$

# ACTIVITY OF INTEGRATION

*SCENARIO (Panic escape).*

The Figure below shows a general situation in which a stream of people attempt to escape through an exit door that turns out to be locked. The people move toward the door at speed ***Vs*= 3.50 m/s**, are each ***d* = 0.25 m** in depth, and are separated by ***L* = 1.75 m**. The arrangement in Figure below occurs at time ***t* = 0**.




(a) At what average rate does the layer of people at the door increase? and what does the answer reveal about the situation

(b) At what time does the layer's depth reach 5.0 m? what does the answer reveal about the situation?

(c) What lessons do you learn from this situation and how can we avoid injuries and loss of lives that may result from such a situation?

# ACTIVITY OF INTEGRATION

You are to drive **300 km** to an interview. The interview is at **11:15 A.M**. You plan to drive at **100 km/h**, so you leave at **8:00 A.M**. to allow some extra time. You drive at that speed for the first **100 km**, but then construction work forces you to slow to **40 km/**h for **40 km**.

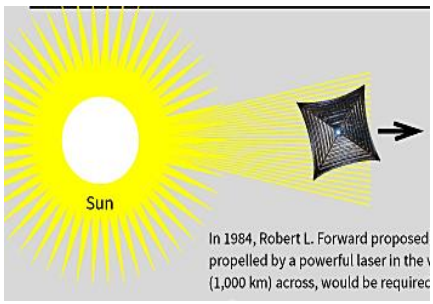


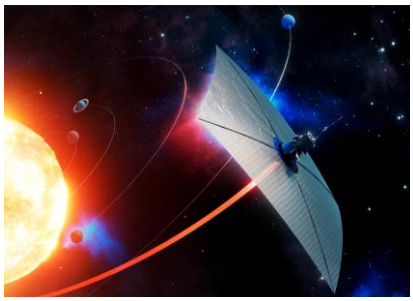
(a) What would be the least speed needed for the rest of the trip to arrive in time for the interview?

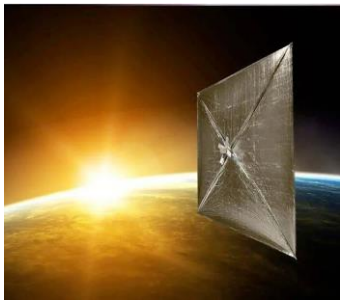
(b) And knowing how dangerous over rushing to beat the deadline can be to your life and other road users, explain how best you can overcome these dangers as you drive on your way to the interview?

# ACTIVITY OF INTEGRATION

A "sun yacht" is a spacecraft with a large sail that is pushed by sunlight. Although such a push is tiny in everyday circumstances, it can be large enough to send the spacecraft outward from the Sun on a cost-free but slow trip. Suppose the **1st model** of the spacecraft has a mass of **900 kg** and receives a push of **20 N**. The company wants to build the **2nd model** which is **faster** than the **1st model**.





(a) If you are the space engineer hired to build it, what **minimum** distance must it travel at in 1 day for it to be faster than the 1st model? (Assuming the spacecraft started from rest)

(b) What **minimum** velocity must the second model travel at if it is to be faster than the 1st model as required by the company?

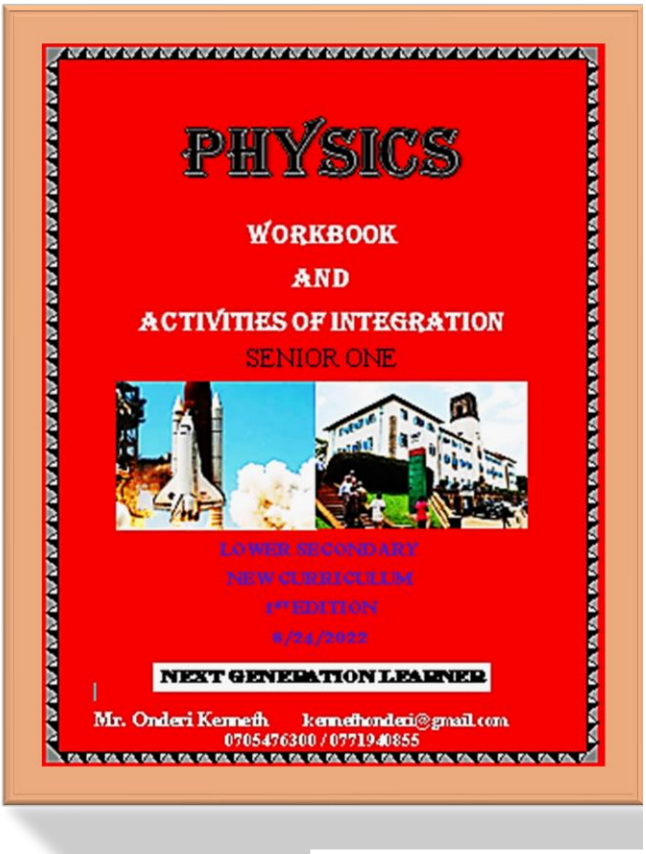
(c) What adjustments need to be made in the 1st model to make the 2nd model a success?

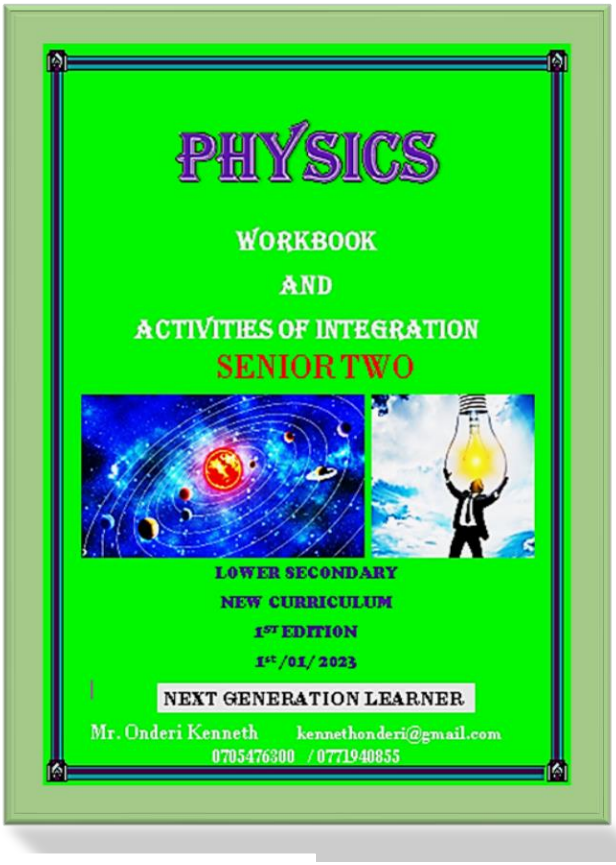
(d) Why is it important for scientists to study about the sun?

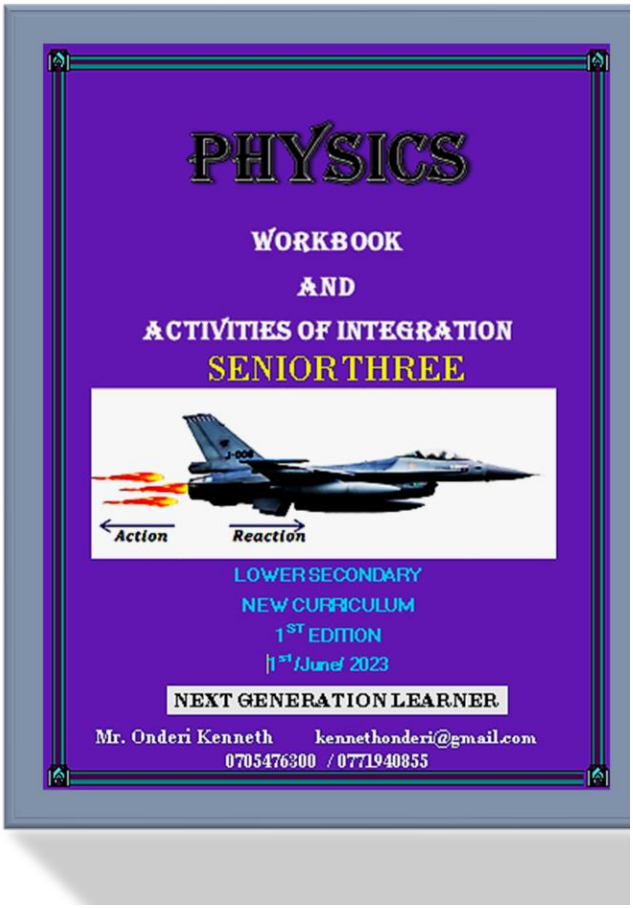
# ACTIVITY OF INTEGRATION

Uganda is country with roads but most of them have potholes. One day two cars were moving to town but in an attempt for the smaller car to overtake the large one, the driver hit a pothole and lost control of the car and knocked the large car in the process damaging both cars. some people lost their lives in the process. Due to many accidents, the local radio station has invited the traffic police inspector to address the community. If you are the traffic police inspector of that area, write down the speech that you will present to the community at the radio station.




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- This book will ease your work as a teacher following competence-based curriculum.
- This will ease the student's work since all types of questions have been included which require knowledge application and research where necessary.
- The questions have been made as simple as possible to enable students easily understand (very many examples, pictures and illustrations)
- The various Activities of integration have also been included at the end of each topic to enable students apply the knowledge attained from the already learnt topic.
- The worksheet is to help students practice as many questions as possible. The space provided is just enough for each question asked.
- The books provide a brief introduction, short answer questions in the right sequence as it is recommended by other textbooks available in the market.

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